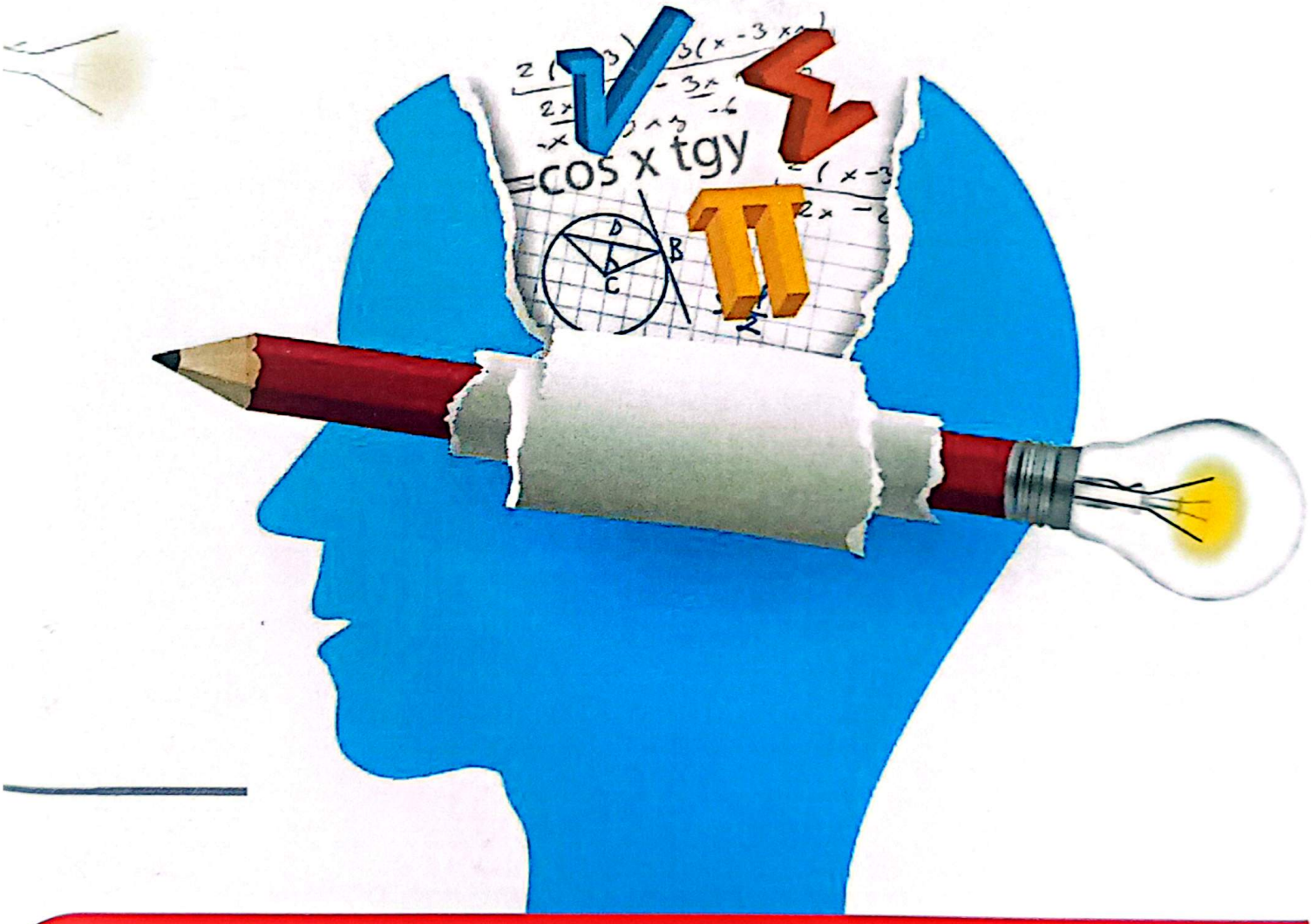
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MATHEMATICS SEMINAR

SUB-MATH ITEM BOOKLET (A-LEVEL)



NEW ADVANCED LEVEL SECONDARY CURRICULUM (UACE)

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SUBSIDIARY MATHEMATICS SEMINAR ITEMS

SECTION A: GEOMETRY AND CALCULUS

Item 1

During preparations for a regional science exhibition, a school technical team sets up three service points on a rectangular compound. The control desk is placed at point A, the camera stand at point B, and the emergency support point is placed at point C. Their positions, measured from the main gate O, are represented by the position vectors

$$\overrightarrow{OA} = 5\mathbf{i} + 2\mathbf{j}, \qquad \overrightarrow{OB} = 4\mathbf{i} + 6\mathbf{j}, \qquad \overrightarrow{OC} = -2\mathbf{i} + 6\mathbf{j}..$$

For safety and clear communication, the team plans to lay two signal cables from the emergency support point, one to the camera stand and the other to the control desk. The cable from C to B should not exceed 7 units, while the cable from C to A should be less than 9 units. The two cable routes should also make an angle greater than $40°$ but less than $80°$, so that people can move safely between the service points.

At the same exhibition, a rotating search light is programmed so that its direction angle x satisfies

$$6\cos^2 x - \cos x - 1 = 0,$$

where $0° \leq x \leq 360°$. The search light will only be used if all possible directions in this interval can be identified correctly.

Task

As a mathematics student, use mathematical justification to help the technical team decide whether:

(a) the available cable materials should be used by checking whether the routes CB and CA satisfy the stated length limits;

(b) the space between the two cable routes should be approved for pedestrian movement by checking whether the angle between CB and CA lies within the safe range;

(c) the search light should be programmed for use by finding all possible direction angles x in the interval $0° \leq x \leq 360°$.

Item 2

A community health team is organising a free medical camp at a trading centre. The main registration tent is taken as the origin O. A triage point P, where patients are first examined, is located with position vector

$$\overrightarrow{OP} = \begin{pmatrix} 2 \\ 3 \end{pmatrix},$$

while a medicine collection point Q is located with position vector

$$\overrightarrow{OQ} = \begin{pmatrix} 4 \\ -1 \end{pmatrix}.$$

The organisers want to mark the three points O, P, and Q to form a triangular movement zone for patients and health workers.

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To avoid long walking distances for sick patients, each side of the triangular movement zone should not exceed 6 units. For proper crowd control, the angle between the routes OP and OQ should be at least $60°$, so that the two queues do not become too close to each other.

At the same event, a rotating emergency lamp is programmed so that its direction angle x satisfies

$$\sin^2 x + \sin x \cos x = 0,$$

where $0° \le x \le 360°$. The lamp will only be used if all possible direction angles in this interval can be identified correctly.

Task

As a mathematics student, use mathematical justification to help the community health team decide whether:

(a) the triangular movement zone OPQ should be marked out by checking whether each side is within the maximum walking-distance limit;

(b) the two patient queues should be allowed to start from O by checking whether the angle between OP and OQ is wide enough for crowd control;

(c) the emergency lamp should be adjusted for use by finding all possible direction angles x satisfying the given trigonometric equation.

Item 3

A community road-safety team is planning three temporary service points for a weekend clean-up and health campaign. Point A is a water station, point B is a waste collection point, and point C is a first-aid point. Their position vectors from the community office O are given by

$$\overrightarrow{OA} = 4\mathbf{i} - \mathbf{j}, \qquad \overrightarrow{OB} = \mathbf{i} + 3\mathbf{j}, \qquad \overrightarrow{OC} = -5\mathbf{i} + 2\mathbf{j}.$$

The team wants to mark the directed routes $\overrightarrow{AB}$, $\overrightarrow{BC}$, and $\overrightarrow{CA}$ so that volunteers know the correct movement direction between the service points. For easy movement, the triangular working area should have all its angles clearly determined, and each angle should be greater than $20°$ so that no corner is too narrow for people carrying equipment.

During the same activity, a ladder is placed against a wall to hang a campaign banner. The angle θ made by the ladder with the ground is acute, and the height-to-ladder ratio is given by

$$\sin\theta = \frac{7}{25}.$$

For safe support, the ladder will be used only if the base-to-ladder ratio is at least 0.90 and the height-to-base ratio is not more than 0.35.

Task

As a mathematics student, use mathematical justification to help the community road-safety team decide whether:

(a) the volunteer movement routes should be painted on the ground by correctly determining the directed vectors $\overrightarrow{AB}$, $\overrightarrow{BC}$, and $\overrightarrow{CA}$;

(b) the triangular working area should be approved by checking whether each angle of triangle ABC is greater than $20°$;

(c) the ladder should be used for hanging the banner by finding $\cos\theta$ and $\tan\theta$, then checking whether the base-to-ladder and height-to-base ratios satisfy the safety limits.

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along the path. The area enclosed by the curve, the vertical lines $x = 0$ and $x = 1$, and the x-axis will be covered with paving slabs.

In another section of the garden, the slope of a second curved path at any point is given by

$$\frac{dy}{dx} = 3x^2 - 2x + 1.$$

The path passes through the point $(-1, 4)$, and its equation is needed for accurate construction.

Task

As a mathematics student, use mathematical justification to help the community youth group decide whether:

(a) the curved pedestrian path should be accepted by finding the turning point of $y = (2+x)(1-x)$ and identifying whether it gives a highest or lowest point;

(b) the number of paving slabs should be estimated by sketching the curve clearly and calculating the area bounded by the curve, $x = 0$, $x = 1$, and the x-axis;

(c) the second curved path should be constructed by finding its equation from the given gradient function and the point $(-1, 4)$.

Item 8

A community recreation group is designing a curved walking track and a drainage edge for a public garden. The amount of material needed for part of the curved track is represented by the integral

$$\int_{-1}^{3} x(x+1)\,dx.$$

For another section of the garden, the gradient function of a curved boundary is given by

$$\frac{dy}{dx} + 4 = 12x.$$

The curve passes through the point $(-2, 30)$, and its equation is needed for setting out the boundary accurately.

At the point on this curve where $x = 1$, a straight drainage edge is to be constructed normal to the curve. The drainage edge will be accepted only if its equation is correctly determined.

Task

As a mathematics student, use mathematical justification to help the community recreation group decide whether:

(a) the material estimate for the curved track should be accepted by evaluating $\int_{-1}^{3} x(x+1)\,dx$;

(b) the curved boundary should be set out by finding its equation from $\frac{dy}{dx} + 4 = 12x$ and the point $(-2, 30)$;

(c) the drainage edge should be constructed by finding the equation of the normal to the curve at $x = 1$.

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SECTION B PART I: ALGEBRA

Item 9

A secondary school is selecting teachers to attend a professional development workshop. The school has 9 A-level teachers. In the Science department, there is one teacher for each of the subjects Mathematics, Physics, Chemistry, and Biology. In the Arts department, there is one teacher for each of the subjects Economics, Geography, History, Literature, and Fine Art.

The school can send only 3 teachers to the workshop. The transport budget can support any one group of 3 teachers, but the administration wants to know the total number of possible groups before making the final selection. For balanced academic representation, the administration will be satisfied only if the probability of selecting at least two Science teachers is greater than $\frac{1}{3}$. The workshop organisers have also stated that the Mathematics teacher must attend; therefore, the school must determine how many possible groups can still be formed under this condition.

In the same school, the academic department is preparing a revision programme whose weekly target follows an arithmetic progression. The sum of the first 16 weekly targets is 1088, and the 16th target is twice the 8th target. The programme will be adopted only if the first weekly target is not less than 8, so that the revision schedule begins with enough work.

Task

As a mathematics student, use mathematical justification to help the school administration decide whether:

(a) the workshop selection list should be prepared by determining the total number of possible groups of 3 teachers;

(b) the selection method should be considered suitable for Science representation by checking whether the probability of selecting at least two Science teachers is greater than $\frac{1}{3}$;

(c) the workshop and revision plans should be adopted by finding the number of teams when the Mathematics teacher must attend and checking whether the first term of the arithmetic progression is at least 8.

Item 10

A school games committee is preparing records for a football tournament and a related sports-day arrangement. In the store, some equipment records are represented by the matrices

$$M = \begin{pmatrix} 4x & 6 \\ -5 & -2x \end{pmatrix}, \qquad N = \begin{pmatrix} -1 & -2 \\ 3 & 3y \end{pmatrix}, \qquad K = \begin{pmatrix} y & 4 \\ -2 & 12 \end{pmatrix}.$$

The committee expects the records to satisfy $K = M + N$. The values of x and y will be accepted only if they are positive and do not exceed 6, so that the equipment record remains within the planned stock limit.

In the football tournament, three teams, Arsenal, Chelsea, and Liverpool, had the following results:

Team	Wins	Draws	Losses
Arsenal	2	1	1
Chelsea	2	0	2
Liverpool	1	2	1

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The teams are awarded 3 points for a win, 1 point for a draw, and no point for a loss. The tournament winner will be confirmed only if the leading team gets at least 7 points.

For the decoration stand, the simplifying factor for one of the measurements is

$$\frac{6\sqrt{10}+2\sqrt{40}}{\sqrt{2}\times\sqrt{20}}.$$

The measurement will be accepted only if the exact simplified value is not greater than 5.

Task

As a mathematics student, use mathematical justification to help the school games committee decide whether:

(a) the equipment record should be accepted by finding x and y from $K = M + N$ and checking whether both values are positive and not greater than 6;

(b) the football tournament winner should be confirmed by using matrix multiplication to calculate the points for each team and checking whether the leading team has at least 7 points;

(c) the decoration measurement should be used by simplifying the surd expression exactly and checking whether its value is not greater than 5.

Item 11

A school welfare committee is planning a termly savings programme for supporting needy learners. The amounts saved each week are arranged in an arithmetic progression. The second weekly saving is 15 units, and the fifth weekly saving is 21 units.

The programme will be adopted only if the common difference is positive but does not exceed 5, so that the weekly increase is manageable. The first weekly saving must also be at least 10 units, and the total amount saved in the first 10 weeks must be at least 200 units.

In another planning activity, the committee uses the equation

$$3\log_t(10+3t) = 6$$

to determine a scale factor t. The scale factor will be accepted only if $t > 1$ and $t \leq 6$.

Task

As a mathematics student, use mathematical justification to help the school welfare committee decide whether:

(a) the weekly savings increase should be approved by finding the common difference and checking whether it is positive and not greater than 5;

(b) the savings programme should be adopted by finding the first term and the sum of the first 10 terms, then checking whether the starting amount and total savings meet the required limits;

(c) the planning scale factor should be accepted by solving $3\log_t(10+3t) = 6$ and checking whether $1 < t \leq 6$.

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SECTION B PART II: DATA ANALYSIS AND PROBABILITY

Item 12

A school academic committee is comparing the performance of 10 students, labelled A to J, in Mathematics and Physics tests. The scores are shown in the table below.

Student	A	B	C	D	E	F	G	H	I	J
Mathematics (x)	28	20	40	28	21	31	36	29	33	24
Physics (y)	30	20	40	28	22	35	35	27	31	23

The committee wants to use the relationship between Mathematics and Physics scores to guide learner support. A student who scores 37 in Physics will be advised on the expected Mathematics performance using a line of best fit. The relationship will be considered useful only if the rank correlation coefficient shows a strong positive association.

In a related agricultural science project, the masses of animals on a school demonstration farm are normally distributed with mean 60 kg and variance 25 kg^2. The farm manager wants to know the percentage of animals weighing at least 52.5 kg. The feeding plan will be maintained only if at least 90% of the animals weigh 52.5 kg or more.

During a school educational tour, a traveller compares the price index for breakfast (B), lunch (L), and supper (S) in Kampala and Mbarara. The price indices are shown below.

Town	Breakfast (B)	Lunch (L)	Supper (S)
Kampala	120	130	125
Mbarara	115	135	110

The actual quantities consumed by the traveller for breakfast, lunch, and supper are 300 g, 400 g, and 300 g, respectively. The school will recommend the cheaper town for meals if its weighted index is lower.

Task

As a mathematics student, use mathematical justification to help the school academic committee, farm manager, and tour organisers decide whether:

(a) learner support should be planned using the scatter diagram by drawing a line of best fit and estimating the Mathematics score of a student who scored 37 in Physics;

(b) the Mathematics–Physics relationship and the farm feeding plan should be accepted by calculating the rank correlation coefficient and checking whether at least 90% of the animals weigh 52.5 kg or more;

(c) the cheaper meal town should be recommended by calculating the weighted price index for Kampala and Mbarara using the given quantities.

Item 13

A local newspaper company is reviewing its weekly sales performance over a period of 12 weeks. The sales, in thousands of copies, are shown in the table below.

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Week	1	2	3	4	5	6	7	8	9	10	11	12
Number of copies sold	315	378	490	430	510	580	565	595	640	660	628	670

The company wants to smooth the weekly sales using 3-week moving averages and then plot both the original sales and the moving averages on the same axes. The manager will increase printing for week 13 only if the graph suggests that expected sales are at least 650 thousand copies.

In the same company, a promotional gift is given to customers using a random variable X, whose probability distribution is

$$P(X = x) = \begin{cases} \frac{x}{10}, & x = 1, 2, 3, \\ 0, & \text{elsewhere.} \end{cases}$$

The promotion will be maintained only if the expected value of X is at least 1.4.

The company also tests 8 trainee sales agents and obtains the marks

$$5,\ 9,\ 11,\ 15,\ 19,\ 15,\ 10,\ 14.$$

The training programme will be considered effective only if the mean mark is at least 12 and the variance is not more than 18.

Task

As a mathematics student, use mathematical justification to help the newspaper company decide whether:

(a) printing should be increased for week 13 by calculating the 3-week moving averages, plotting them with the original sales, and checking whether the expected sales are at least 650 thousand copies;

(b) the promotional gift system should be maintained by finding $P(1 \leq X < 3)$ and checking whether $E(X) \geq 1.4$;

(c) the trainee sales programme should be accepted by calculating the mean and variance of the marks, then checking whether the mean is at least 12 and the variance is not more than 18.

Item 14

A factory produces animal food packed in bags. The weights of the bags are normally distributed with mean 50 kg and standard deviation 2.8 kg. For quality control, the factory wants to know the probability that a randomly selected bag weighs more than 52 kg, lies between 46 kg and 55 kg, and weighs less than 54 kg. The packaging process will be accepted only if the percentage of bags weighing less than 54 kg is at least 90%.

In a school students' council, there are 7 girls and 5 boys. Two students are selected at random to represent the council at a meeting. The selection process will be considered fair only if the probabilities of selecting two girls and selecting a boy first followed by a girl can be determined clearly.

In another institution, the enrolment of students over five years is shown below.

Year	2003	2004	2005	2006	2007
Number of students	145	182	170	155	213

The institution uses three-year moving averages to monitor enrolment trends. It will plan for expansion in 2008 if the enrolment for 2008, found using the fourth moving average of 203, is more than 230 students.

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Task

As a mathematics student, use mathematical justification to help the factory, students' council, and institution decide whether:

(a) the packaging process should be approved by finding the required normal-distribution probabilities and checking whether at least 90% of the bags weigh less than 54 kg;

(b) the council selection process should be considered clear and fair by finding the probabilities of selecting two girls and selecting a boy first followed by a girl;

(c) the institution should plan for expansion in 2008 by calculating the three-year moving averages and checking whether the projected enrolment is more than 230 students.

Item 15

A farm manager is studying the weights, in kilograms, of 50 cattle on a farm. The weights recorded are:

60	81	76	68	84	112	76	102	86	67
65	98	107	110	72	99	87	92	76	77
94	102	87	86	73	118	98	120	62	87
65	92	104	116	91	93	78	122	102	92
80	111	73	120	106	123	94	109	80	96

The manager wants to group the data using equal class intervals starting with $60 - 69$, and then draw an ogive to study the distribution of cattle weights. The feeding programme will be reviewed if the median weight is below 90 kg, or if fewer than 10 cattle weigh 118 kg and above.

In another section of the farm, it was observed that 3 seeds in every 4 seeds planted germinate. If 16 seeds are planted, the farm manager wants to estimate the expected number of seeds that will germinate and the probability that exactly 14 seeds will germinate. The planting method will be maintained if the expected number of seeds germinating is at least 12.

The farm also supports a university student whose expenditures in 2005 and 2006 are shown below.

Item	2005	2006	Weight
Text books	100000	120000	3
Pocket money	50000	70000	2
Research	40000	50000	1

Using 2005 as the base year, the sponsor wants to calculate the weighted aggregate price index. The student allowance will be increased if the weighted aggregate price index is greater than 120.

Task

As a mathematics student, use mathematical justification to help the farm manager and sponsor decide whether:

(a) a grouped frequency table and an ogive can be formed from the cattle weights, and whether the median weight and the number of cattle weighing 118 kg and above satisfy the feeding-programme conditions;

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(b) the expected number of seeds that will germinate and the probability that exactly 14 seeds will germinate can be determined, and whether the planting method should be maintained;

(c) the weighted aggregate price index can be calculated using 2005 as the base year, and whether the student allowance should be increased.

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